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Market Feedback: Evidence from the Horse's Mouth

Contents

1	Overview	3
2	Setting, data, and survey design	3
2.1	Institutional setting	3
2.2	Survey waves and questions	3
2.3	Who answers the survey? (Figure 1)	4
3	Direct survey evidence (Figures 2–4) and immediate implications	4
3.1	Prevalence of monitoring (Figure 2)	4
3.2	Motives / channels (Figure 3)	4
3.3	What information do firms claim to learn? (Figure 4)	4
3.4	Correlations across channels and content (Table 2)	5
4	Conceptual framework and hypotheses	5
4.1	Hypothesis 1: Who reports learning? (determinants of Learn)	5
4.2	Hypothesis 2: Do firms act on learning? (validation)	5
4.3	Hypothesis 3: Active management of trading suspensions	5
5	Empirical models (all specifications in the paper)	5
5.1	Notation and variable construction	5
5.2	Model 1: Determinants of the learning channel (Eq. (1))	6
5.3	Model 2: Investment-to-price sensitivity (Eq. (2))	6
5.4	Model 3: Learning and M&A performance (Eq. (3))	7
5.5	Model 4: Trading suspensions (Eq. (4))	7
6	Results	7
6.1	Figure 1: Respondent positions	8
6.2	Figure 2: Monitoring prevalence (Question I)	8
6.3	Figure 3: Channels (Question II)	8
6.4	Figure 4: Information content (Question III, 2022)	9
6.5	Table 1: Summary statistics	9

6.6	Table 2: Correlations between responses	9
6.7	Table 3: Testing Hypothesis 1 (who reports learning?)	9
6.8	Table 4: Testing Hypothesis 2(a) (investment-to-price sensitivity)	10
6.9	Table 5: Testing Hypothesis 2(b) (content-specific validation)	11
6.10	Table 6: Testing Hypothesis 2(c) in M&A (value implications)	11
6.11	Table 7: Trading suspension facts	11
6.12	Table 8: Trading suspensions and channels (Hypothesis 3)	12
6.13	Appendix Tables A1–A4	12
7	Key takeaways	13
8	Conclusion	13
8.1	Summary	13
8.2	Main conclusions	14
8.3	Why the cross-section looks the way it does	15
8.4	Intuition for the validation tests (investment, M&A, suspensions)	15
8.5	Takeaway	15

1 Overview

The paper provides *direct* evidence on *market feedback*—the idea that financial markets affect real corporate decisions—by surveying (nearly) the universe of Chinese publicly listed firms. Two surveys (2019 and 2022) ask firms (i) whether they monitor their own and peers’ stock prices, (ii) *why* they do so (learning vs. financing vs. other motives), and (iii) *what information* they think stock prices contain (macro/industry, policy/regulation, competitive position, etc.). The authors then validate the survey answers by linking them to observable corporate actions: (a) learning firms exhibit higher *investment-to-price sensitivity* (a canonical prediction of informational feedback), (b) firms that say they learn about M&A opportunities conduct more acquisitions and achieve better post-merger outcomes, and (c) firms manage trading suspensions in a way consistent with their reported channel (learning vs. financing), especially around large price drops.

2 Setting, data, and survey design

2.1 Institutional setting

The surveys are administered through the CSRC electronic system to listed firms on the Shanghai and Shenzhen exchanges. A practical advantage is extremely high response rates, greatly reducing non-response concerns.

2.2 Survey waves and questions

Let i index firms.

- **2019 survey:** Questions I and II.
- **2022 survey:** Questions I, II, and III (the information-content module).

Question I (Prevalence): “How does your company pay attention to the stock market?” Choices distinguish whether the firm monitors (a) its own stock price, (b) peers’ stock prices, (c) both, (d) only an index, or (e) not at all.

Question II (Channels): Conditional on monitoring, “Which of the following is the reason that your company pays attention to the stock price of your own company?” Choices map into *channels*:

- **Learning channel (Learn):** price contains new information useful for real investment decisions;
- **Financing channel (Fin):** price affects refinancing / external financing conditions;
- plus additional motives (compensation/option payoffs, monitoring/pressure, M&A protection).

Question III (Information content, 2022 only): Conditional on choosing the learning channel, respondents rate (Likert scale) whether stock prices are useful for learning about: macro/industry, policies/regulations, competitive position, customer demand, technology, cost of capital, M&A opportunities, COVID impact, and a “no information” statement.

2.3 Who answers the survey? (Figure 1)

Firms report the respondent’s position; most answers are prepared by board secretaries (a key executive role in China), with a non-trivial share from chairperson/CEO/CFO/other executives. This matters because it addresses a typical concern for survey-based corporate finance: are respondents informed and engaged?

3 Direct survey evidence (Figures 2–4) and immediate implications

3.1 Prevalence of monitoring (Figure 2)

Across both waves, the vast majority of firms monitor *both* their own and peers’ stock prices. The fraction that does not monitor the stock market at all is very small (and smaller in 2022 than in 2019). Importantly, the pattern is similar across respondent ranks, which supports internal consistency of responses.

3.2 Motives / channels (Figure 3)

Two motives dominate:

1. **Learning** (stock prices contain information new for real decisions);
2. **Financing** (stock prices affect refinancing conditions).

Other motives (compensation, monitoring pressure, takeover protection) exist but are clearly second-order in frequency. The stability of these frequencies across respondent ranks again supports the view that survey answers are meaningful.

3.3 What information do firms claim to learn? (Figure 4)

Among learning firms, the *most strongly affirmed* information dimensions are:

- macro/industry conditions,
- policies and regulations,
- competitive position.

Moderate affirmation appears for customer demand, technology developments, cost of capital, and M&A opportunities; COVID information is weaker. A key “sanity check” item is the “no information” statement, which receives extremely low agreement.

3.4 Correlations across channels and content (Table 2)

Correlations between channel indicators (Learn, Fin, etc.) are small in magnitude. Interpretation: these are *distinct* motives rather than one latent “attention” factor. Information-content indicators are positively correlated (macro and policy in particular), consistent with the idea that learning is multi-dimensional yet structured.

4 Conceptual framework and hypotheses

The paper’s empirical work is guided by a stylized informational feedback model (developed in the Online Appendix) that maps *signal precision* into predicted reliance on prices.

4.1 Hypothesis 1: Who reports learning? (determinants of Learn)

Let Learn_i be a dummy for choosing the learning channel in Question II. The model motivates the following comparative statics:

Hypothesis 1 (Explaining the learning channel). A firm is more likely to report learning from its stock price if: (a) investors’ information precision is higher; (b) managerial private information precision is lower; (c) analysts’ information precision is lower (empirically ambiguous, but predicted under a “analysts transmit managerial info” view); (d) managerial sophistication is higher; and/or (e) the firm’s stock price is more informative.

4.2 Hypothesis 2: Do firms act on learning? (validation)

Survey claims should map into real actions and outcomes:

Hypothesis 2 (Validating learning). (a) Learning firms exhibit higher investment-to-price sensitivity. (b) Firms affirming learning about specific information types exhibit higher investment-to-price sensitivity. (c) Learning improves investment performance / shareholder value (tested in M&A settings).

4.3 Hypothesis 3: Active management of trading suspensions

Trading suspensions in China provide a revealed-action margin.

Hypothesis 3 (Trading suspensions). (a) Learning firms suspend trading less frequently (they *want* informative prices). (b) Financing-motivated firms suspend more when prices drop sharply (to maintain prices for financing).

5 Empirical models (all specifications in the paper)

5.1 Notation and variable construction

- Firms: i , time: t (year), m (month).

- Survey-based channel indicators: Learn_i , Fin_i .
- Investment: $\text{Capxrnd}_{i,t}$ (capex + R&D scaled by assets).
- Valuation: $Q_{i,t}$ (Tobin's Q).
- Controls in investment regressions: cash flow $\text{CF}_{i,t}$, future return $\text{Ret3}_{i,t}$, inverse assets $\text{InvAst}_{i,t}$.
- M&A outcomes: MA is one of NAnnounce , NCompletion , BHAR , CAR .
- Trading suspension: $\text{Susp}_{i,m}$ ("important matters" reason), and $\text{PriceDrop}_{i,m}$ (bottom-quartile monthly return).

5.2 Model 1: Determinants of the learning channel (Eq. (1))

The baseline cross-sectional Probit specification is:

$$\text{Learn}_i = a + b \cdot \text{Factor}_i + c' \cdot \text{Controls}_i + \varepsilon_i, \quad (\text{Probit; report marginal effects}). \quad (1)$$

Key object: coefficient b varies with the chosen Factor proxy set.

Factor blocks (Table 3).

- **Investor information:** institutional ownership (InsShares), short-selling eligibility (Short).
- **Managerial private information:** insider trading intensity (Insider), earnings surprise proxy (ERC).
- **Analyst information:** number of analysts (NAnalysts), number of forecasts (NForecasts).
- **Managerial sophistication:** professional background share (Professional), education level (Degree).
- **Price informativeness:** $1 - R^2$, adjusted PIN (AdjPIN), price delay (PriceDelay).

5.3 Model 2: Investment-to-price sensitivity (Eq. (2))

The panel specification (firm-year) is:

$$\text{Capxrnd}_{i,t+1} = \alpha_i + \beta_t + c \cdot Q_{i,t} + d' \cdot \text{Controls}_{i,t} + u_{i,t}. \quad (2)$$

Validation logic: Compare $\hat{c}$ across subsamples defined by survey answers:

- Learn vs. NoLearn (Table 4) $\Rightarrow$ Hypothesis 2(a).
- Affirm vs. NoAffirm for each information type in Question III (Table 5) $\Rightarrow$ Hypothesis 2(b).

5.4 Model 3: Learning and M&A performance (Eq. (3))

Let LearnAcq_i measure the degree to which a firm claims to learn M&A information from prices (constructed from Question III). Then:

$$MA = a + b \cdot \text{LearnAcq} + g' \cdot \text{Controls} + \eta, \quad (3)$$

where MA is alternately $N\text{Announce}$, $N\text{Completion}$, $BHAR$, or CAR . Fixed effects depend on the outcome (see Table 6 caption in the paper).

5.5 Model 4: Trading suspensions (Eq. (4))

At the firm-month level:

$$\text{Susp}_{i,m} = \beta_{t(m)} + c \cdot (\text{Feedback}_i \times \text{PriceDrop}_{i,m}) + d \cdot \text{Feedback}_i + e \cdot \text{PriceDrop}_{i,m} + h' \cdot \text{Controls}_i + \xi_{i,m}, \quad (4)$$

where Feedback_i is either Learn_i or Fin_i . The authors estimate Probit models and report marginal effects. The interaction term operationalizes Hypothesis 3(b): financing-motivated firms should suspend more when prices drop.

6 Results

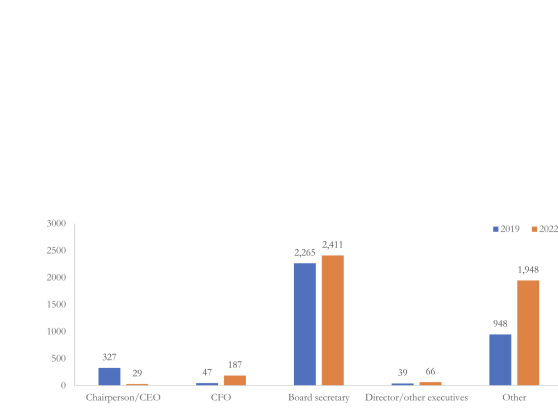


Figure 1. Distribution of respondents' positions in their firms. This figure plots the distribution of the positions of the respondents that were assigned by their firms to respond to our 2019 and 2022 market feedback surveys. 3,626 Chinese public firms listed on the Shanghai and Shenzhen Stock Exchanges responded to the 2019 survey, and 4,641 firms responded to the 2022 survey.

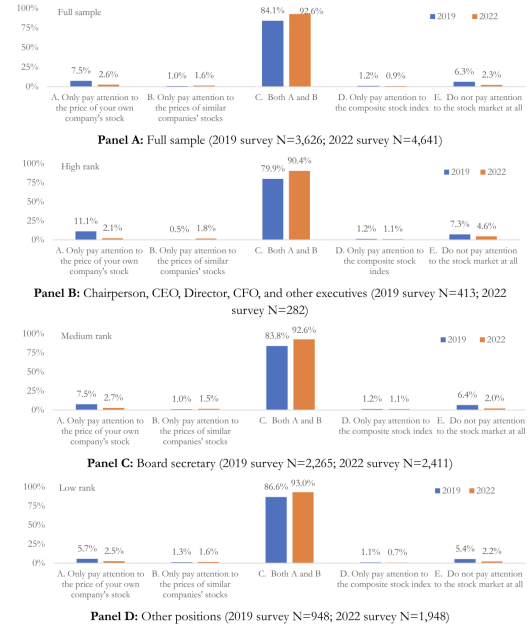


Figure 2. Responses to Question 1. This figure plots the frequencies for each choice by the responding firms in survey Question 1 ("How does your company pay attention to the stock market?"). Panel A presents results in the full sample. Panel B, C, and D present results in high-, medium-, and low-ranking respondents, respectively.

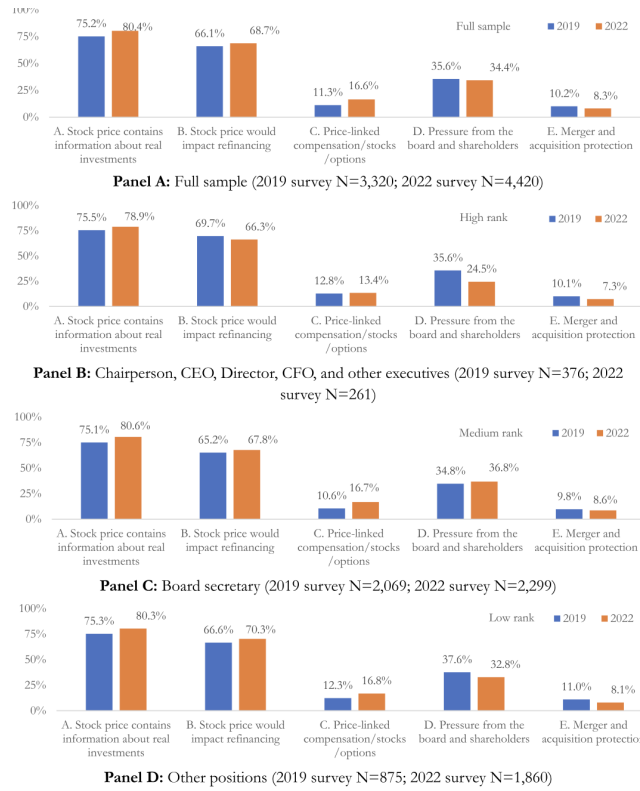


Figure 3. Responses to Question II. This figure plots the frequencies for each choice by the responding firms in Question II (“If you choose A or C in I: Which of the following is the reason that your company pays attention to the stock price of your OWN company?”). Panel A presents results in the full sample. Panel B, C, and D present results in high-, medium-, and low-ranking respondents, respectively.

6.1 Figure 1: Respondent positions

What it shows: Most responses are prepared by board secretaries; meaningful shares are from executives (chairperson/CEO/CFO) and other related functions. **Why it matters:** It supports the credibility of responses: board secretaries handle disclosure and investor relations; executives observe real decisions.

6.2 Figure 2: Monitoring prevalence (Question I)

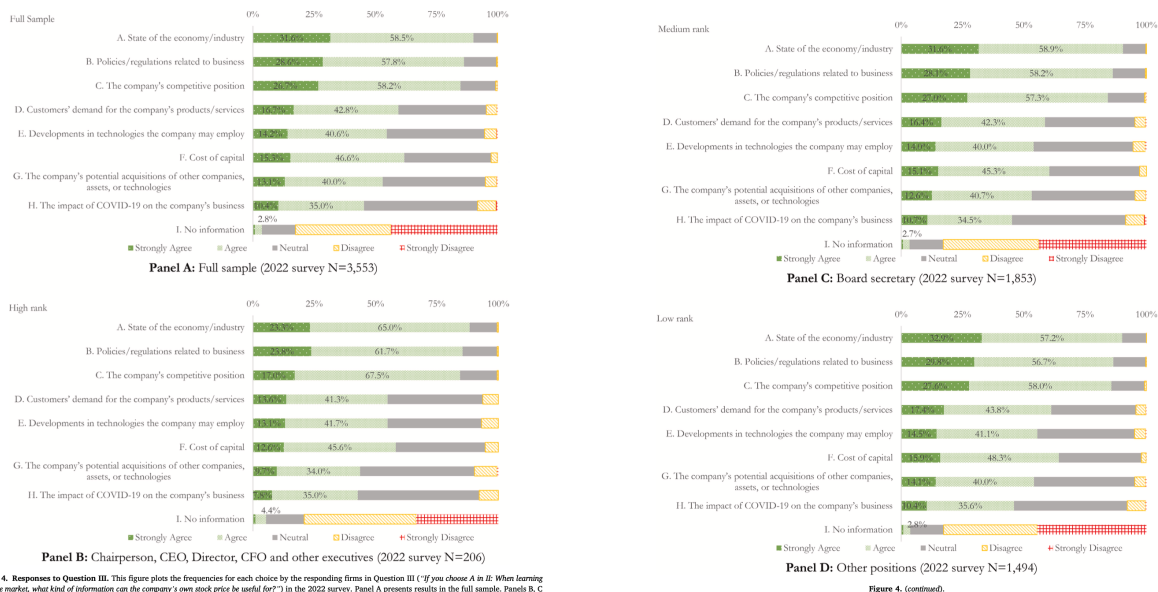
Main pattern: Monitoring is nearly universal; the dominant choice is “monitor both own and peers’ prices.” **Interpretation:** In a broad sample of firms, market prices are a salient object in managers’ information set.

6.3 Figure 3: Channels (Question II)

Main pattern: Learning and financing dominate; other channels are secondary. **Interpretation:** Market feedback is not a niche phenomenon; it is mostly about (i) extracting information and (ii) external finance conditions.

6.4 Figure 4: Information content (Question III, 2022)

Main pattern: Strongest agreement for macro/industry, policy/regulation, and competitive-position information; moderate agreement for demand/technology/cost-of-capital/M&A; weak for “no information.” **Interpretation:** The claimed informational content is largely *external* to the firm and plausibly aggregated in prices.



6.5 Table 1: Summary statistics

What it contains: Means/medians/SDs for firm characteristics for 2019 and 2022 respondent samples (effectively the population of listed firms). **Useful reading:**

- The listed-firm sector has moderate leverage and non-trivial institutional ownership.
- Analyst coverage is sizable on average (but skewed: median smaller than mean).
- Short-selling eligibility increases substantially by 2022, consistent with market development.

6.6 Table 2: Correlations between responses

Panel A/B (channels): Low correlations across channel dummies $\Rightarrow$ firms have distinct reasons for monitoring; learning is not mechanically the same as financing or monitoring pressure. **Panel C (content):** Information-content indicators are positively correlated, especially macro with policy, consistent with macro/policy co-movement.

6.7 Table 3: Testing Hypothesis 1 (who reports learning?)

Table 3 estimates Eq. (1) with different Factor proxies.

Table 1

Summary statistics for the responding firms.

This table reports summary statistics for the 3,626 Chinese public firms responding to the 2019 survey, and the 4,641 firms responding to the 2022 survey. Firm information is as of 2018 for the 2019 survey, and is as of 2021 for the 2022 survey, respectively.

	2019 Survey (N = 3,626)			2022 Survey (N = 4,641)		
	Mean	Median	STD	Mean	Median	STD
Firm Age (year)	20.14	20.05	5.01	21.84	21.74	5.41
Total Assets (billion RMB)	11.71	4.12	18.82	12.57	3.89	21.29
Market Cap. (billion RMB)	7.56	4.07	8.22	13.38	6.43	16.47
Capital Expenditure (%)	4.75	3.12	4.84	5.26	3.40	5.41
R&D Expense (%)	2.21	1.87	1.81	2.80	2.39	2.23
ROA (%)	4.91	5.20	6.79	5.47	5.48	6.76
Tobin's Q	1.80	1.50	0.91	2.62	2.09	1.65
Leverage (%)	43.52	42.26	20.22	42.66	41.35	20.83
No. Analysts	7.59	2.00	11.01	6.66	1.00	10.59
Short Indicator	0.27	0.00	0.44	0.48	0.00	0.50
Insider Trading (%)	0.14	0.00	0.31	0.02	0.00	0.06
Institutional Ownership (%)	37.47	38.20	23.03	35.58	35.64	23.35
1-R ²	0.52	0.50	0.18	0.79	0.84	0.16
SOE	0.32	0.00	0.47	0.30	0.00	0.46

Table 2

Correlations between responses.

This table reports the correlations between channels of market feedback based on responses to Question II ("If you choose A or C in I: Which of the following is the reason that your company pays attention to the stock price of your OWN company?") and the correlations between information contents firms expect to learn about based on responses to Question III ("If you choose A in II: When learning from the market, what kind of information can the company's own stock price be useful for?"). The channel indicators (Learn, Finance, Compensation, Monitor, and M&A) equal one if they are chosen by the respondent, and zero otherwise. The information content indicators (Macro, Policy, Competition, Demand, Technology, Cost, Acquisition, Covid) take the value of 2 if the respondent chooses "strongly agree" for the corresponding information type, 1 for "agree", 0 for "neutral", -1 for "disagree", and -2 for "strongly disagree". ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Market feedback channels (2019 Survey, N = 3,320)								
	Learn	Finance	Compensation	Monitor	M&A			
Learn	1.0000		-0.0313**	-0.1141***	0.0153			
Finance		1.0000	0.0560***	0.0632***	0.0977***			
Compensation			1.0000	0.1229***	0.1668***			
Monitor				1.0000	0.1581***			
M&A					1.0000			
Panel B: Market feedback channels (2022 Survey, N = 4,420)								
	Learn	Finance	Compensation	Monitor	M&A			
Learn	1.0000	-0.0812***	-0.0083	0.1432***	0.0327**			
Finance		1.0000	0.1587***	0.0241***	0.0504***			
Compensation			1.0000	0.1860***	0.1340***			
Monitor				1.0000	0.1525***			
M&A					1.0000			
Panel C: Information contents for learning firms (2022 Survey, N = 3,533)								
	Macro	Policy	Competition	Demand	Technology	Cost	Acquisition	Covid
Macro	1.0000	0.7644***	0.0911***	0.4166***	0.3979***	0.3825***	0.3211***	0.2921***
Policy		1.0000	0.5987***	0.4444***	0.3962***	0.3487***	0.3038***	0.2761***
Competition			1.0000	0.5372***	0.4728***	0.4307***	0.3889***	0.3343***
Demand				1.0000	0.6633***	0.4935***	0.4395***	0.4348***
Technology					1.0000	0.4935***	0.4669***	0.4669***
Cost						1.0000	0.5695***	0.4111***
Acquisition							1.0000	0.5555***
Covid								1.0000

Investor information. Institutional ownership and short-selling eligibility enter positively and significantly. *Interpretation:* If investors are more informed (or can trade on negative information), prices contain more incremental information for managers, increasing the propensity to report learning.

Managerial private information. Insider trading and the earnings-surprise proxy enter negatively and significantly. *Interpretation:* When managers already possess more private information, they rely less on prices and are less likely to claim that prices contain *new* information.

Analyst information. Analyst coverage / forecast measures enter negatively and significantly. *Interpretation:* Consistent with the view that analysts help transmit managerial information to markets (reducing investors' private-information role), thereby lowering incremental informativeness for managers.

Managerial sophistication. Professional background and education are positive (at least weakly) and significant. *Interpretation:* More sophisticated managers are more capable of extracting signals from prices.

Price informativeness. $1 - R^2$ is positive and significant; price delay is negative and significant; adjusted PIN is not precisely estimated. *Interpretation:* When prices incorporate more firm-specific information and do so more quickly, managers are more likely to report learning.

6.8 Table 4: Testing Hypothesis 2(a) (investment-to-price sensitivity)

Table 4 estimates Eq. (2) and compares $\hat{c}$ across subsamples.

- Full sample: Q loads positively and significantly on next year's investment.

Table 3
Information, firm characteristics, and the learning channel.
This table reports the **Investment-valuation Reg**, **Index** of the learning channel following Eq. (1). The sample consists of 4,171 firms responding to the 2022 survey. The dependent variable is a dummy variable that equals one if a firm chooses A in Question II in the 2022 survey, and zero otherwise. The independent variables of interest are investor, managerial, analyst information, managerial sophistication, and price information measures. The respondent position, industry, and province fixed effects are included. See **Appendix A2** for variable definitions. Marginal effects are reported in parentheses are adjusted for heteroscedasticity and clustering at the respondent position level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Investor Info.				Managerial Info.				Analyst Info.				Mng. Sophistication				Price Information			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Age																				
Age	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Size																				
Size	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Profit																				
Profit	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Industry																				
Industry	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Province																				
Province	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Constant																				
Constant	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Adjusted R²																				
Adjusted R ²	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)

Table 4

Investment-to-price sensitivity conditional on whether to learn.
This table reports the OLS regression results about firm investments following Eq. (2). The **Full** sample includes 4,171 responding firms in the 2022 survey. The **Learn** (**NoLearn**) subsample includes firms selecting (not selecting) the learning channel in Question II in the 2022 survey. The sample period is from 2012 to 2021. The dependent variable is capital expenditure plus R&D expenses. The firm and year fixed effects are included. See **Appendix A2** for definitions of variables. Standard errors reported in parentheses are adjusted for heteroscedasticity and clustering at the firm level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Y = Capexrnd	(1)	(2)	(3)
	Full	Learn	NoLearn
Q	0.0030*** (0.0006)	0.0038*** (0.0007)	0.0007 (0.0011)
CF	0.0741*** (0.0106)	0.0674*** (0.0121)	0.0963*** (0.0213)
Ret3	-0.0043*** (0.0009)	-0.0042*** (0.0010)	-0.0048** (0.0020)
InvAst	17.6647*** (3.6817)	16.9093*** (4.4597)	19.3833*** (6.2245)
Cons.	0.0382*** (0.0020)	0.0369*** (0.0024)	0.0430*** (0.0038)
FEs	Yes	Yes	Yes
N	16,156	12,273	3,883
R ²	0.5454	0.5498	0.5356
Diff in Coef.			0.0031**

- **Learn firms:** the Q coefficient is larger and highly significant.
- **NoLearn firms:** the Q coefficient is small and statistically insignificant.

Interpretation: Firms that say they learn behave in a way consistent with informational feedback: investment responds more to valuation signals.

6.9 Table 5: Testing Hypothesis 2(b) (content-specific validation)

Table 5 partitions learning firms into **Affirm** vs. **NoAffirm** for each information type in Question III and re-estimates Eq. (2). **Main pattern:** Q coefficients are systematically larger in the Affirm subsample; differences are significant for nearly all content types (except the COVID item). **Interpretation:** The survey module is not just “cheap talk”; firms that claim to learn a particular type of information display stronger investment responses to prices.

6.10 Table 6: Testing Hypothesis 2(c) in M&A (value implications)

Table 6 estimates Eq. (3) using LearnAcq.

- Firms that claim to learn M&A information announce and complete more deals.
- Conditional on completion, post-merger abnormal returns and announcement CARs are higher.

Interpretation: In a setting where decision quality is observable in outcomes, learning from markets is associated with value creation rather than mere activity.

6.11 Table 7: Trading suspension facts

What it shows: Suspensions are not rare; the “important matters” category accounts for a large share and has meaningful duration. **Why it matters:** “Important matters” suspensions are sufficiently discretionary to be a choice variable, making them informative about firms’ motives.

Table 5
Investment-to-price sensitivity conditional on what to learn about.

This table reports the OLS regression results about firm investments following Eq. (2). For each information type in Question III, a firm falls into the *Affirm* subsample if it chooses “strongly agree” or “agree”; otherwise, the firm falls into the *NoAffirm* subsample. The sample period is from 2012 to 2021. The dependent variable is capital expenditure plus R&D expenses. The firm and year fixed effects are included. See Appendix A2 for definitions of variables. Standard errors reported in parentheses are adjusted for heteroscedasticity and clustering at the firm level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Y = Capital	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Info =	Macro	Micro	Policy	Competition	Acquisition	Financing	Other	Other
Sample =	Affirm	NoAffirm	Affirm	NoAffirm	Affirm	NoAffirm	Affirm	NoAffirm
Q	0.0041*** (0.0008)	0.0004 (0.0009)	0.0044*** (0.0008)	0.0006 (0.0009)	0.0044*** (0.0008)	0.0005 (0.0009)	0.0044*** (0.0008)	0.0011 (0.0009)
CP	0.0039*** (0.0013)	0.0029*** (0.0017)	0.0022*** (0.0013)	0.0028*** (0.0013)	0.0038*** (0.0013)	0.0039*** (0.0013)	0.0039*** (0.0013)	0.0022*** (0.0013)
Rec3	-0.0047*** (0.0011)	-0.0035*** (0.0017)	-0.0046*** (0.0011)	-0.0034*** (0.0011)	-0.0044*** (0.0011)	-0.0035*** (0.0011)	-0.0044*** (0.0011)	-0.0033*** (0.0011)
InvAr	16.6759*** (4.5353)	19.1900*** (5.9928)	14.1341*** (4.6952)	23.0766*** (5.6693)	16.2964*** (4.8752)	19.5645*** (5.3121)	21.7721*** (5.9489)	14.6705*** (4.3146)
Cons.	0.0258*** (0.0025)	0.0435*** (0.0034)	0.0368*** (0.0025)	0.0415*** (0.0034)	0.0357*** (0.0025)	0.0431*** (0.0031)	0.0322*** (0.0025)	0.0431*** (0.0034)
FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	11,940	5,116	10,591	5,565	10,405	5,751	11,778	8,978
R ²	0.0547	0.0294	0.0485	0.0543	0.0492	0.0542	0.0482	0.0465
Diff in Coef.	0.0039***		0.0038***		0.0039***		0.0043***	
Y = Capital	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Info =	Technology	Cost of Capital	Acquisition	Financing	Other	Other	Other	Other
Sample =	Affirm	NoAffirm	Affirm	NoAffirm	Affirm	NoAffirm	Affirm	NoAffirm
Q	0.0051*** (0.0011)	0.0017*** (0.0007)	0.0044*** (0.0009)	0.0019*** (0.0006)	0.0048*** (0.0010)	0.0018*** (0.0007)	0.0042*** (0.0011)	0.0024*** (0.0007)
CP	0.0171*** (0.0176)	0.0162*** (0.0131)	0.0166*** (0.0118)	0.0175*** (0.0143)	0.0171*** (0.0121)	0.0171*** (0.0118)	0.0171*** (0.0128)	0.0168*** (0.0128)
Rec3	-0.0046*** (0.0015)	-0.0042*** (0.0011)	-0.0046*** (0.0011)	-0.0049*** (0.0011)	-0.0049*** (0.0011)	-0.0052*** (0.0011)	-0.0049*** (0.0011)	-0.0049*** (0.0011)
InvAr	17.6897*** (5.9677)	17.8215*** (4.5414)	14.4373*** (5.8220)	20.4397*** (4.6259)	17.7030*** (5.5420)	12.8295*** (5.8807)	19.8795*** (4.6638)	17.8215*** (4.6638)
Cons.	0.0333*** (0.0033)	0.0411*** (0.0025)	0.0380*** (0.0033)	0.0383*** (0.0025)	0.0423*** (0.0024)	0.0369*** (0.0025)	0.0389*** (0.0025)	0.0389*** (0.0025)
FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	6,422	9,734	7,381	8,775	6,493	5,507	10,649	5,507
R ²	0.0501	0.0411	0.0547	0.0377	0.0388	0.0521	0.0500	0.0437
Diff in Coef.	0.0033***		0.0024**		0.0030**		0.0017	

Table 6

The learning channel and merger performance.

This table reports the OLS regression results on the effects of learning on firms' merger performance following Eq. (3). The sample consists of M&A transactions from 2018 to 2022 in which firms responding to the 2022 survey are acquirers. The dependent variables include the number of M&A announcements, the number of completed mergers, the buy-and-hold abnormal return during the 12 months following the completion, and the cumulative abnormal return from day -1 to day 1 around the announcement. The independent variable is the degree to which a manager learns information about potential M&A opportunities according to the 2022 survey. The respondent position, industry, and province fixed effects are included in columns (1) and (2); and the year fixed effects are further included in columns (3) and (4). See Appendix A2 for variables definitions. Standard errors reported in parentheses are adjusted for heteroscedasticity and clustering at the industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Y=	(1)	(2)	(3)	(4)
	NAnnounce	NCompletion	BHAR	CAR
LearnAcq	0.0531*** (0.0158)	0.0167** (0.0063)	0.0140*** (0.0033)	0.0011** (0.0005)
Size	0.1412*** (0.0267)	0.0252* (0.0136)	-0.0075 (0.0098)	0.0001 (0.0013)
Leverage	0.9316*** (0.2012)	0.4857*** (0.0482)	0.1575*** (0.0222)	0.0016 (0.0049)
History	0.0241*** (0.0037)	0.0050*** (0.0013)	-0.0014* (0.0007)	0.0002 (0.0002)
SOE	-0.1246 (0.0788)	-0.0830*** (0.0200)	-0.0038 (0.0134)	0.0019 (0.0014)
Constant	-0.4585** (0.1609)	0.0553 (0.1046)	0.0099 (0.0767)	0.0001 (0.0100)
FEs	Yes	Yes	Yes	Yes
Observations	4,166	4,166	1,997	5,093
R ²	0.0900	0.0337	0.0512	0.0241

6.12 Table 8: Trading suspensions and channels (Hypothesis 3)

Table 8 estimates Eq. (4) with Feedback_i being Learn or Fin.

- **Learn:** negative marginal effect on suspensions (learning firms suspend less).
- **Learn $\times$ PriceDrop:** positive (learning firms become more likely to suspend after very negative returns).
- **Fin:** weak in level, but **Fin $\times$ PriceDrop** is positive and significant (financing-motivated firms suspend more when prices fall sharply).

Interpretation: The revealed-action margin is consistent with the survey channels: learning firms generally prefer trading to continue (so prices remain informative), while financing firms act to stabilize prices when they drop.

6.13 Appendix Tables A1–A4

- **A1–A3:** Industry variation in responses (monitoring, channel choices, and content) shows broad prevalence with meaningful heterogeneity (e.g., learning less prevalent in some financial industries).
- **A4:** Variable definitions for the empirical sections (useful as a replication checklist).

Table 7
Summary of trading suspensions.
This table reports summary statistics for trading suspensions by firms responding to the 2022 survey from 2020 to 2022.

Reason	Full sample		> 1 day (4 trading hours)	
	N. Suspension	Duration (trading hours)	N. Suspension	Duration (trading hours)
All	1532	11.8	1047	17.1
- Important matters	791	16.5	788	16.6
- Transaction related	484	6.8	10	21.6
- Major risk	130	6.4	130	6.4
- M&A/restructure	105	24.1	105	24.1
- Financing/Shareholder meeting/Media report	8	18.5	8	18.5
- Unknown/others	14	9.7	6	20.7

Table 8

Real effects and trading suspensions.

This table reports the Probit regression results about the effects of the learning and financing channels on firms' trading suspensions following Eq. (4). The sample covers trading suspensions by firms responding to the 2022 survey from 2020 to 2022. The dependent variable is a dummy variable indicating whether a firm suspends the trading of its stock in a month. The independent variables of interest include dummy variables indicating whether the firm reports the learning/financing channels in our survey. The year, month, position, industry, and province fixed effects are included. See Appendix A2 for definitions of variables. Marginal effects are reported. Standard errors reported in parentheses are adjusted for heteroscedasticity and clustering at the year level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

$Y = \text{Susp}$	(1)	(2)	(3)	(4)
$\text{Feedback} =$	<i>Learn</i>		<i>Fin</i>	
<i>Feedback</i>	-0.0012*** (0.0004)	-0.0015*** (0.0005)	0.0000 (0.0005)	-0.0002 (0.0006)
<i>PriceDrop</i>		-0.0012 (0.0010)		-0.0012 (0.0010)
<i>Feedback*PriceDrop</i>		0.0011*** (0.0002)		0.0012*** (0.0004)
<i>Size</i>	-0.0005 (0.0004)	-0.0005 (0.0004)	-0.0005 (0.0004)	-0.0005 (0.0004)
<i>Leverage</i>	0.0038*** (0.0005)	0.0038*** (0.0005)	0.0038*** (0.0006)	0.0038*** (0.0005)
<i>History</i>	0.0002*** (0.0000)	0.0002*** (0.0000)	0.0002*** (0.0000)	0.0002*** (0.0000)
<i>SOE</i>	0.0002 (0.0003)	0.0002 (0.0003)	0.0003 (0.0003)	0.0002 (0.0002)
<i>FES</i>	Yes	Yes	Yes	Yes
<i>N</i>	133,463	133,463	133,463	133,463
<i>Pseudo R²</i>	0.0438	0.0441	0.0428	0.0431

7 Key takeaways

1. **Survey evidence is not the endpoint:** the paper uses survey answers to generate *testable predictions* and validates them using actions (investment, M&A, suspensions).
2. **Learning is about external, aggregative information:** macro/policy/competition dominate, consistent with prices aggregating dispersed information.
3. **Channel separation is essential:** learning and financing imply different behavior (especially in suspensions).
4. **Methodological template:** “ask” (survey) → “validate” (behavioral regression) → “triangulate” (an orthogonal revealed-action test).

8 Conclusion

8.1 Summary

This paper concludes that **market feedback is pervasive in practice:** managers of public firms actively monitor stock-market information and treat it as an input into real corporate decisions. Surveying (essentially) all Chinese public firms in 2019 and 2022, the authors document that **over 90%** of firms say they monitor the stock market, and the two dominant motives managers report are:

1. **Learning:** managers believe prices contain *new* information relevant for real decisions;
2. **Financing:** managers depend on prices for external funding conditions.

The paper then focuses on the learning motive and concludes that (i) the cross-section of who reports learning follows the logic of an information-aggregation model, and (ii) reported learning is behaviorally meaningful, because it predicts real actions and outcomes (investment, M&A, and trading-suspension choices).

8.2 Main conclusions

The findings form a consistent chain:

1. **Managers say they learn from prices, and they can articulate what they learn.** Among learning firms, the most strongly affirmed information dimensions are **macro/industry conditions**, **policy/regulation**, and **competitive position**, i.e., information that is plausibly dispersed across market participants and valuable to aggregate.
2. **Reported learning is strongest when prices are most likely to carry incremental information.** Firms are more likely to report learning when investor information is stronger (e.g., higher institutional ownership; short-selling eligibility), and less likely to report learning when managerial private information is stronger (e.g., insider-trading intensity; earnings-surprise proxies) or when analyst information is stronger (e.g., analyst coverage/forecasting intensity). Learning is also more common when managers are more sophisticated and when the stock price appears more informative.
3. **Learning firms behave as if they use price signals: they exhibit higher investment-to-price sensitivity.** In firm-year panel regressions, the sensitivity of next-year investment (capex + R&D) to Tobin’s Q is substantially higher for firms that report learning, and weak/insignificant for firms that do not.
4. **The validation is content-specific: the investment response is strongest precisely for firms that affirm learning about particular information types.** When the sample is split by whether managers affirm learning about (say) policy or competition from their own price, the Q -investment sensitivity is larger in the affirming subsample, strengthening the interpretation that survey responses reflect real decision rules.
5. **Learning about acquisition opportunities predicts M&A activity and M&A performance.** Firms that claim to learn acquisition-relevant information from prices announce and complete more deals, and exhibit better post-deal abnormal performance measures.
6. **Managers actively manage the trading process in a way consistent with their stated channel.** Using trading suspensions as a revealed-action test, learning firms are less likely to suspend (consistent with valuing price informativeness), while financing-motivated firms are more likely to suspend when prices drop sharply (consistent with “defend-the-price” incentives).

8.3 Why the cross-section looks the way it does

The learning channel is most naturally interpreted through Bayesian updating: managers combine their private information with the information embedded in prices. A minimal representation is:

$$\text{Manager posterior} = \frac{\tau_M s_M + \tau_P s_P}{\tau_M + \tau_P}, \quad \text{weight on price signal} = \frac{\tau_P}{\tau_M + \tau_P},$$

where s_M is the manager's private signal with precision τ_M and s_P is the price-based signal with precision τ_P . This immediately yields the paper's comparative statics:

- If **investor information precision** rises (higher τ_P), managers optimally put more weight on prices and are more likely to report learning.
- If **managerial private information precision** rises (higher τ_M), prices add less incremental information, so learning is less likely to be reported.
- If **prices are more informative** (faster incorporation; more firm-specific content), then s_P is a better signal, raising the value of learning.
- If **manager sophistication** is higher, the effective ability to extract s_P from prices is higher, again increasing learning.

The paper's empirical patterns match these predictions, supporting the interpretation that managers treat prices as signals rather than as mere sentiment.

8.4 Intuition for the validation tests (investment, M&A, suspensions)

Investment. If a manager believes Q contains incremental information about fundamentals or future opportunities, then investment should respond more strongly to Q . Hence higher Q -investment sensitivity is a natural revealed-action signature of market feedback via learning.

M&A. If managers use market prices to infer acquisition opportunities (e.g., overvaluation/undervaluation, rival strength, sector conditions), then learning firms should (i) attempt acquisitions more often and (ii) achieve better outcomes conditional on doing so.

Trading suspensions. Suspensions disrupt information aggregation and therefore reduce price informativeness. Learning firms, which value informative prices, should avoid suspensions. Financing-oriented firms may prefer suspensions during large price drops to reduce price pressure and protect financing conditions. The paper's suspension results align with this channel separation.

8.5 Takeaway

Surveying essentially all Chinese public firms, Goldstein–Liu–Yang show that managers overwhelmingly monitor stock prices and frequently cite learning as the main motive; reported learning varies systematically with the firm's information environment and is validated by real behavior—higher Q -investment sensitivity, better M&A outcomes for firms that say they learn acquisition information,

and trading-suspension choices consistent with learning vs financing incentives—providing direct, large-scale evidence for market feedback.